

RESOLVED & VERIFIED

The Optimizer Reward Numbers Are Fixed

A plain-language report on the bug you found, exactly what we changed, and the safeguards now in place so it can't come back.

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In one paragraph

You noticed the optimizer promising an unbelievable reward total — the Scotiabank Gold American Express showing **500,000 points** for \$100,000 of spending, as if its rich earn rate had no ceiling. It does have a ceiling. We added the real, published spending limits for the cards, fixed the same class of problem in two related calculators, and built a layer of automatic checks that make an impossible number impossible to ship again. Everything has been tested and verified, including the exact card you flagged.

The problem you saw

The optimizer's job is to tell a user which card earns the most on a given purchase. For most everyday amounts it was correct. But for large amounts it produced figures that were too good to be true — the kind of number that makes a careful user stop trusting the whole product.

The clearest example was the one you hit: the Scotiabank Gold Amex earns 5× points on groceries, so the tool simply multiplied $\$100,000 \times 5 = 500,000$ points. In reality that card only pays the boosted rate on the first \$50,000 of spending each year; after that it drops to the basic rate.

Why it happened (no jargon)

Think of a coupon that says "5% back — up to \$50." The app knew the "5% back" part but never recorded the "up to \$50" part. So when someone spent a lot, it kept applying

the bonus forever instead of stopping at the limit.

This wasn't a maths mistake — the calculations were right. It was **missing information**: the reward rates were in the system, but the spending caps that go with them were not. That gap affected a large share of the card catalogue, not just one card.

What we changed

1. An immediate safety net

First, a conservative rule was put in so the tool can *never* show an unbounded number again. If a card's real limit isn't known, it assumes a sensible cautious limit and clearly labels the figure as an estimate. This deliberately under-promises rather than over-promises.

2. The real fix: we added the actual published limits

We researched each card's official terms and entered the genuine spending caps — including the Scotiabank Gold Amex's \$50,000-per-year limit. Where several categories on one card share a single combined limit (common with Scotiabank and Amex cards), that shared limit is now modelled correctly. Every limit we entered is recorded with the source it came from, so it's auditable later.

3. We fixed the same weakness in two sibling tools

The "Buy Points" calculator could recommend buying an impossible quantity of points (programs cap how many you can buy per year), and the "Stack" calculator could project an impossible discount on a big purchase. Both now use the real per-program and per-offer limits.

4. We corrected a related timing bug

For yearly limits, the tool was only looking at the current month's spending instead of the whole year so far. It now measures against the correct period — the full year for annual caps, the month for monthly ones.

The card you flagged: before vs. after

Scotiabank Gold Amex — \$100,000 grocery spend	Before	After
Points the optimizer projects	500,000	300,000
How it's explained to the user	(no explanation — just a big number)	"\$50,000 at 5× + \$50,000 at 1× (cap: \$50,000/yr)"
Reflects the card's real terms?	No	Yes

300,000 = the first \$50,000 at the 5× rate, plus the remaining \$50,000 at the basic 1× rate — exactly what the card actually pays. Verified live against the running application.

How we made sure this won't happen again

This is the part that matters most. The fix isn't just "we corrected the data once." We built standing safeguards that catch this whole family of problems automatically, before anything reaches users.

- ✓ **A built-in rule that can't be switched off.** A non-negotiable check now lives in the code: a card can never be shown earning more than its capped amount allows, for *any* spending figure. If a future change ever breaks this, the check fails loudly.
- ✓ **Automatic tests on every category and spend level.** A test suite runs dozens of combinations — small spends, the exact cap amount, just over the cap, and extreme amounts — and confirms the number is always within the honest limit. The exact Scotiabank Gold scenario you reported is locked in as its own permanent test.
- ✓ **A live end-to-end check.** A script drives the real running optimizer with a range of spending amounts and confirms the live answers stay within bounds — proving it works in practice, not just in theory.
- ✓ **All 14 Pro tools were stress-tested.** Every Pro tool was hit with hostile and nonsensical inputs (zero, negative, one hundred million dollars, broken requests). None produce an impossible projection, none crash, and none leak internal technical details in error messages.
- ✓ **Every "source" link was checked.** A script visits every source/affiliate link the app shows users and reports any that are dead, so broken links can be cleaned up deliberately rather than discovered by a customer.
- ✓ **The actual pages and buttons were tested in a browser.** An automated browser opens the Pro tools, clicks through all four sections and the underlying tiles, and confirms nothing shows a broken screen or a nonsense value — plus the core sign-up and optimizer flows.
- ✓ **A full quality gate before anything ships.** The entire codebase is compiled, type-checked, linted, unit-tested, and the website is production-built; the database changes were proven reversible. Everything passed.

Net effect: for an impossible reward number to reach a user again, it would have to slip past a fixed rule, a wide test matrix, a live check, and a browser test — all of which would fail visibly first. The failure mode you experienced is now actively guarded, not just patched.

What still uses the cautious estimate — and why that’s correct

Some cards genuinely have *no* spending cap (for example PC Optimum cards, Tangerine’s chosen-category cash back, and several Amex and CIBC cards whose terms explicitly say “no limit”). A few others are discontinued products with no current published terms to verify.

For these, the tool keeps using the conservative, clearly-labelled estimate. This is the honest choice: it never *over-*promises, and for truly uncapped cards the estimate only trims numbers that were implausibly large anyway. Each of these cards is documented with the reason, so the list is auditable rather than a mystery.

Other issues we found and fixed along the way

What we noticed	What we did
A second, stale copy of the database existed locally and could mislead testing.	Identified the real database the app uses and documented the trap so future work doesn’t get fooled by it.
Two existing browser tests were failing for an unrelated reason (the landing page had been redesigned).	Updated them so the safety net stays green and trustworthy.
A handful of “source” links were dead, and some sites block automated checks (false alarms).	Produced a clear live/dead report; flagged the genuine dead links for a deliberate clean-up (we did not silently delete data).
A status-tracker tool relied on data arriving in a specific order.	Made it sort the data itself so it’s correct regardless, with a test to prove it.

Bottom line

The credibility problem is resolved.

The optimizer now reflects what the cards actually pay, the two related calculators are fixed the same way, and a durable set of automatic safeguards keeps it that way. The specific Scotiabank Gold Amex case you reported is fixed and permanently tested. Nothing was committed or deployed without your say-so — the work is ready for your review.

Appendix — what changed, in plain terms

Area	Change
Card data	Two new database updates: real per-card spending caps, and real per-program / per-offer purchase & credit limits (each with its source).
Optimizer & calculators	Use the real limits; correct yearly-vs-monthly period; cautious estimate kept only as a labelled fallback.
Safeguards	New permanent tests for the cap rule, a live end-to-end check, a Pro-tools stress script, a link-checker, and browser tests covering all 14 Pro tools.
Documentation	An auditable checklist of every card's cap decision, the updated audit, and a known-issues note so this is never re-diagnosed from scratch.